

Experiment :

Parallel and series Combination of Helical spring.

Objective :

To study the relationship of spring connected in series and parallel and determine the equivalent spring constant.

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Experiment Name: The harmonic Oscillations of Helical spring Parallel and series combination of spring

Objective :-

To study the relationship of spring connected in series and parallel and determined the equivalent spring constant.

Back ground :-

Spring are devices that can store and release energy because of these properties springs are very important in engineering.

Springs can be combined in series and parallel and in a combination of series and parallel. Each spring can be characterized by its spring constant k . The spring constant can be determine by using Hook's law:

$$F = -kx$$

Where;

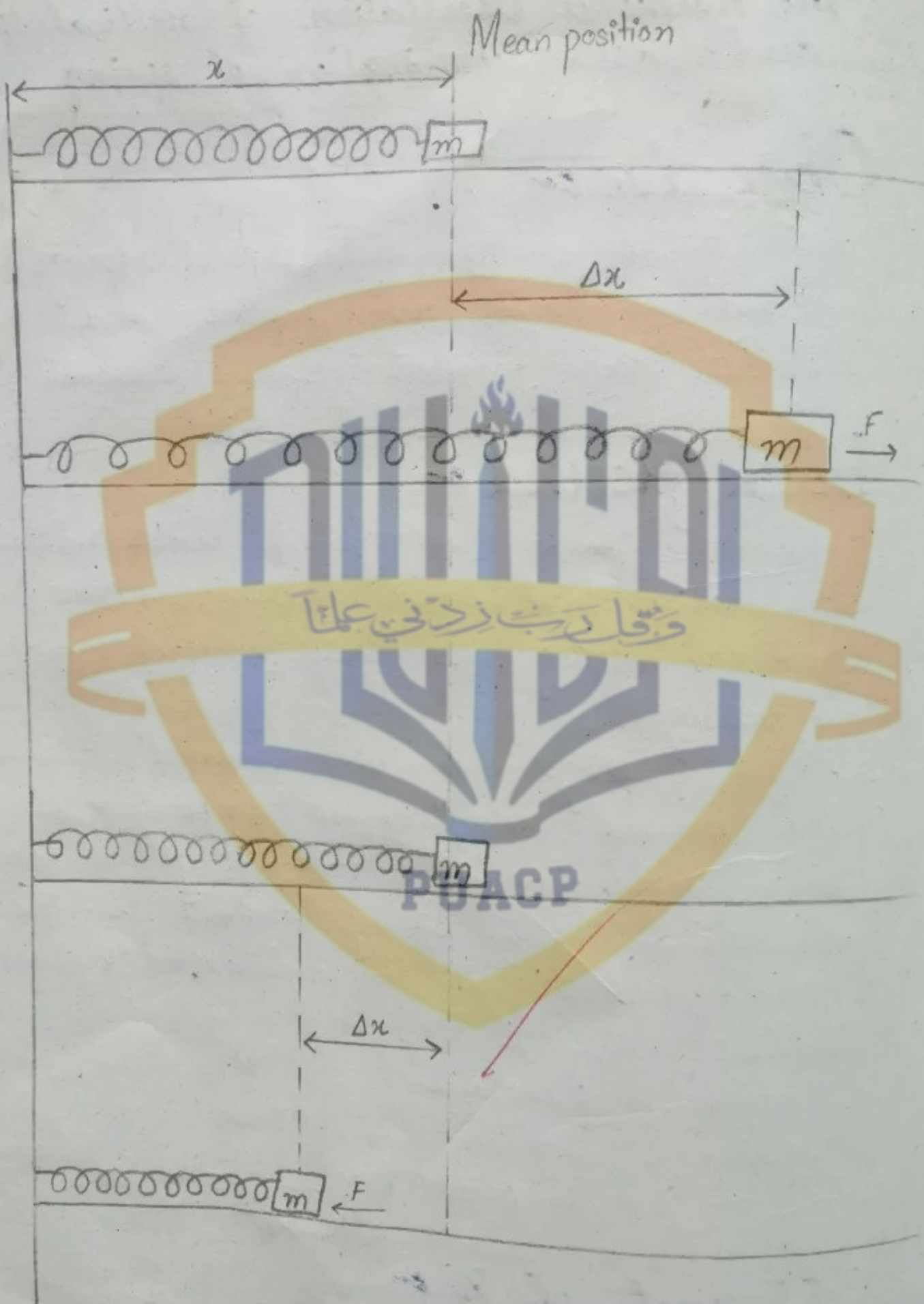
F = applied force

x = resultant displacement

SI unit of k :-

N/m or kg/s^2

Experiment :-



- Weight holder
- Hooks
- slotted weight in grams
- stand with graduated scale
- Helical Spring with pointer
- Rod with Hooks

Theory :-

Theory of experiment is as follow:

Spring constant :-

A characteristic of a spring which is defined as;

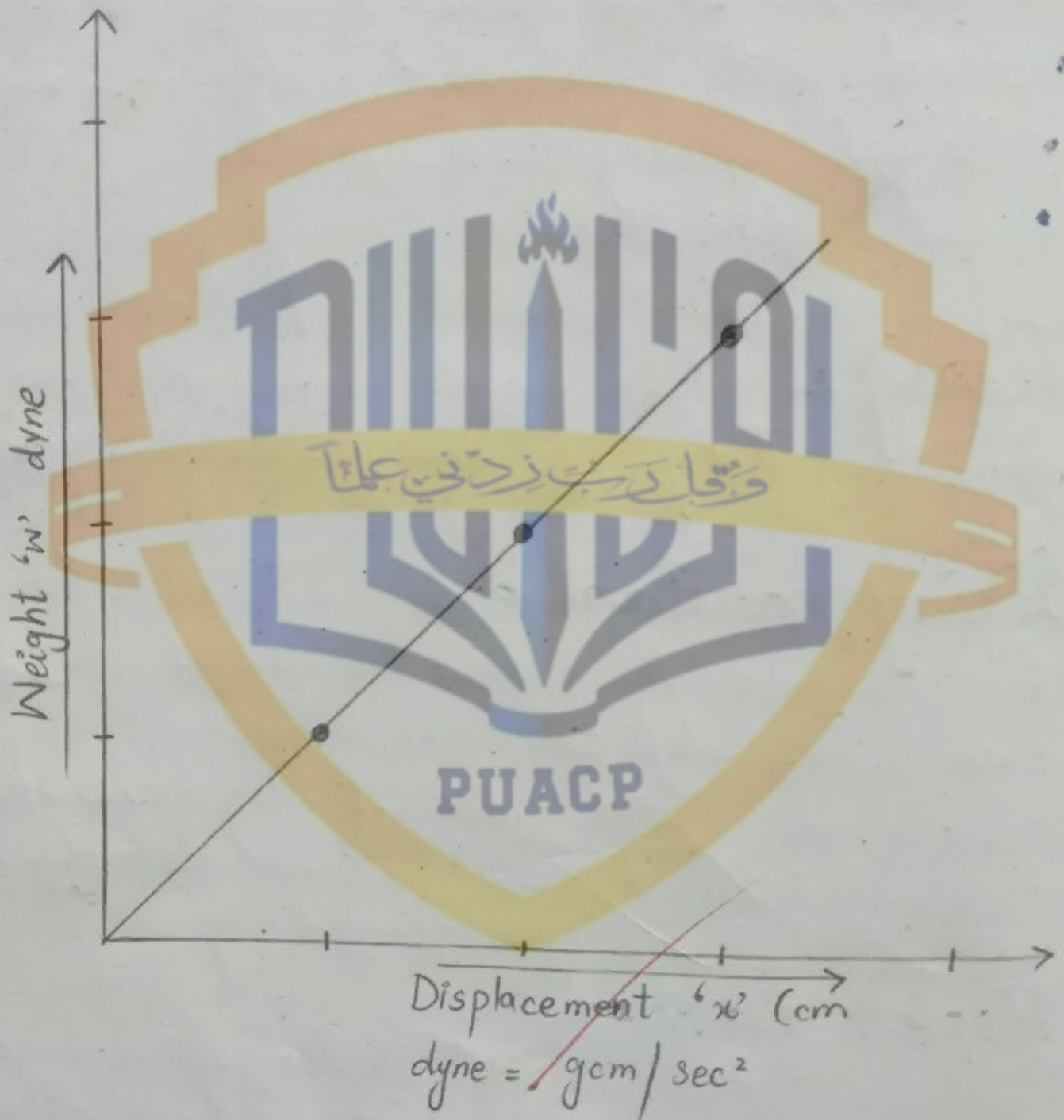
“the ratio of the force affecting the spring to the displacement caused by it.”

Hook's Law :-

“The force F needed to extend compress a spring by some distance x is proportional to the distance.”

$$F = -Kx$$

Where k is constant and negative sign show that displacement is directed toward position where x is the size of



displacement and proportionality constant k is specific for every spring.

Hook's Law and graph :-

Displacement is measure in centim and weight in gram. the weight hang has no mass. The downward pull of gravity is balanced by upward pull of spring. A free or force body diagram of mass spring system is;

$$F = W = kx$$

weight is mass times the acceleration of gravity.

$$W = mg$$

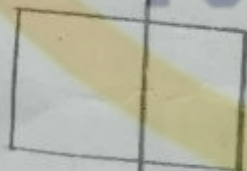
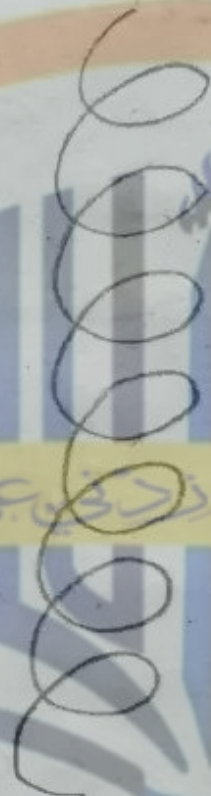
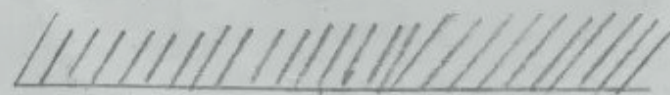
where g is about 9.8 cm/sec^2 . using this relationship k can be calculated as;

$$mg = kx$$

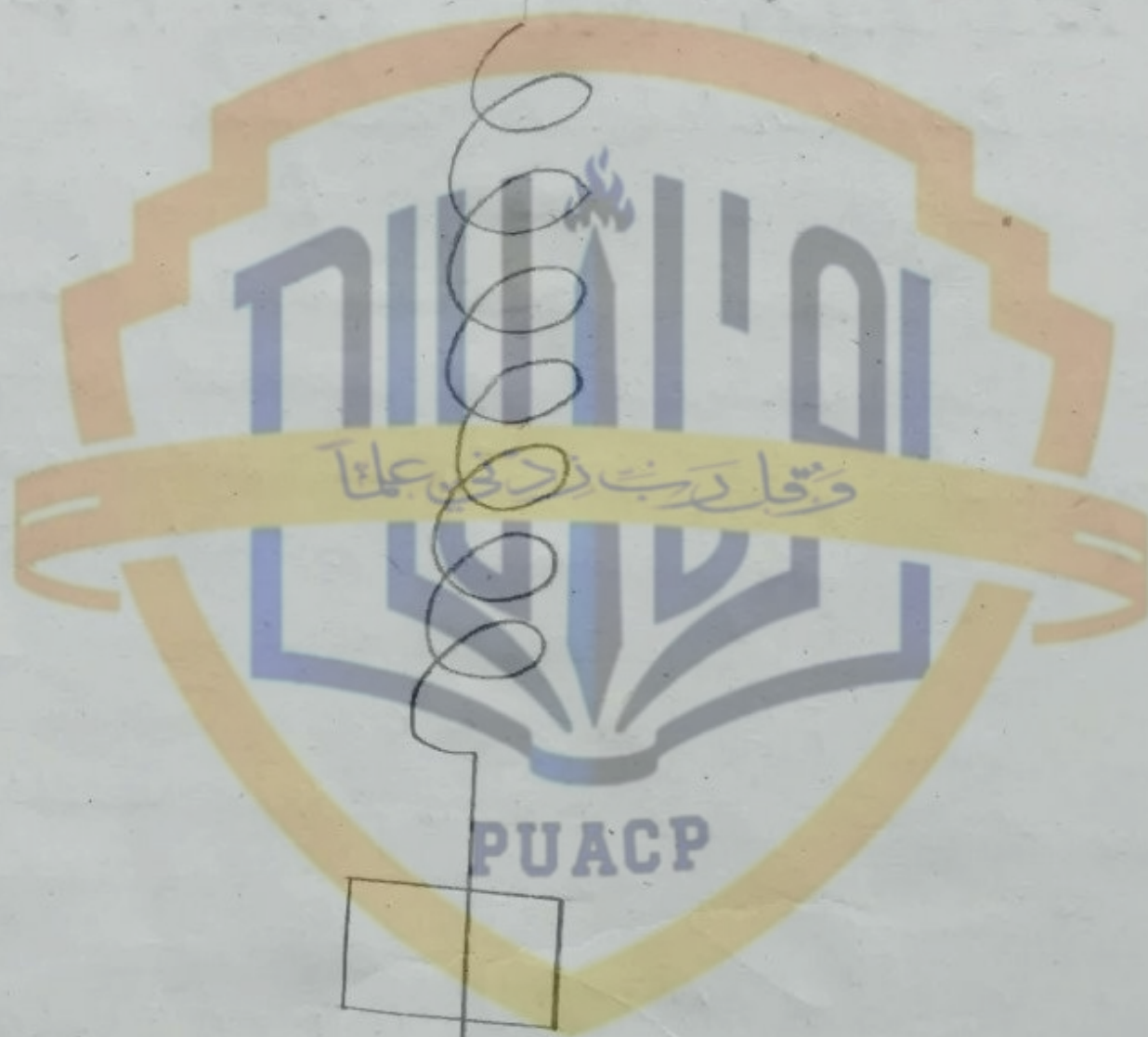
$$k = \frac{mg}{x}$$

$$k = \frac{W}{x}$$

there is a linear relationship b/w w



$W = mg$



and x and w versus x plot a straight line.

Therefore the spring constant k is the slope of straight line w versus x plot.

Restoring force :-

"The force which tends to move the body back to its original position when applied force is removed."

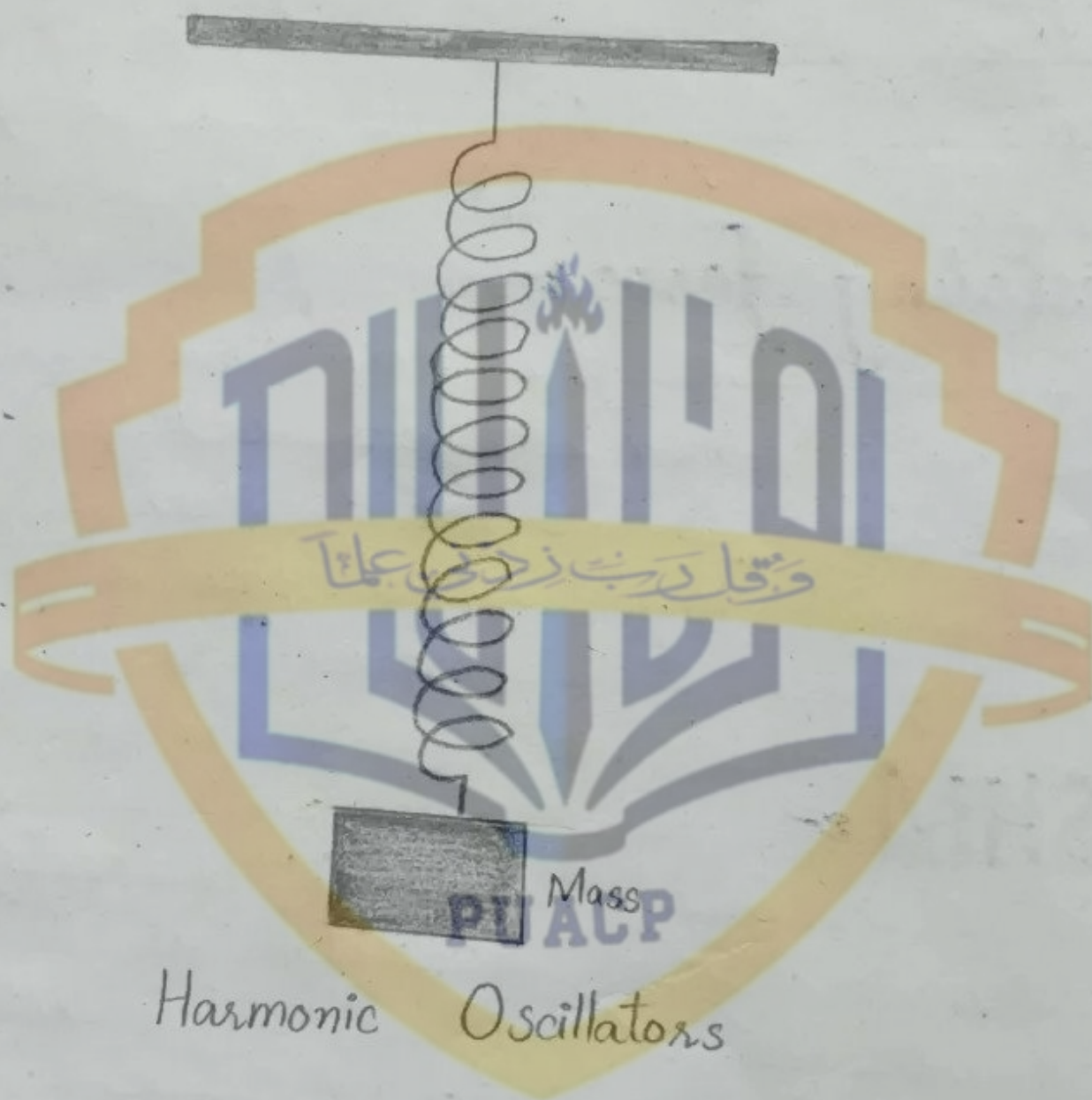
SHM :-

Such a type of vibration motion in which acceleration at any instant of a body executing SHM is proportional to displacement and is always directed towards mean position is called SHM.

Example :-

Mass - Spring System

Instantaneous Displacement :-



Harmonic Oscillators

“The value of the distance of a vibrational body from mean position at any instant is known as instantaneous displacement.”

$$x = x_0 \sin \omega t$$

Amplitude of vibration :-

The maximum displacement of vibrating body on either side from its mean position is called amplitude.

Vibration :-

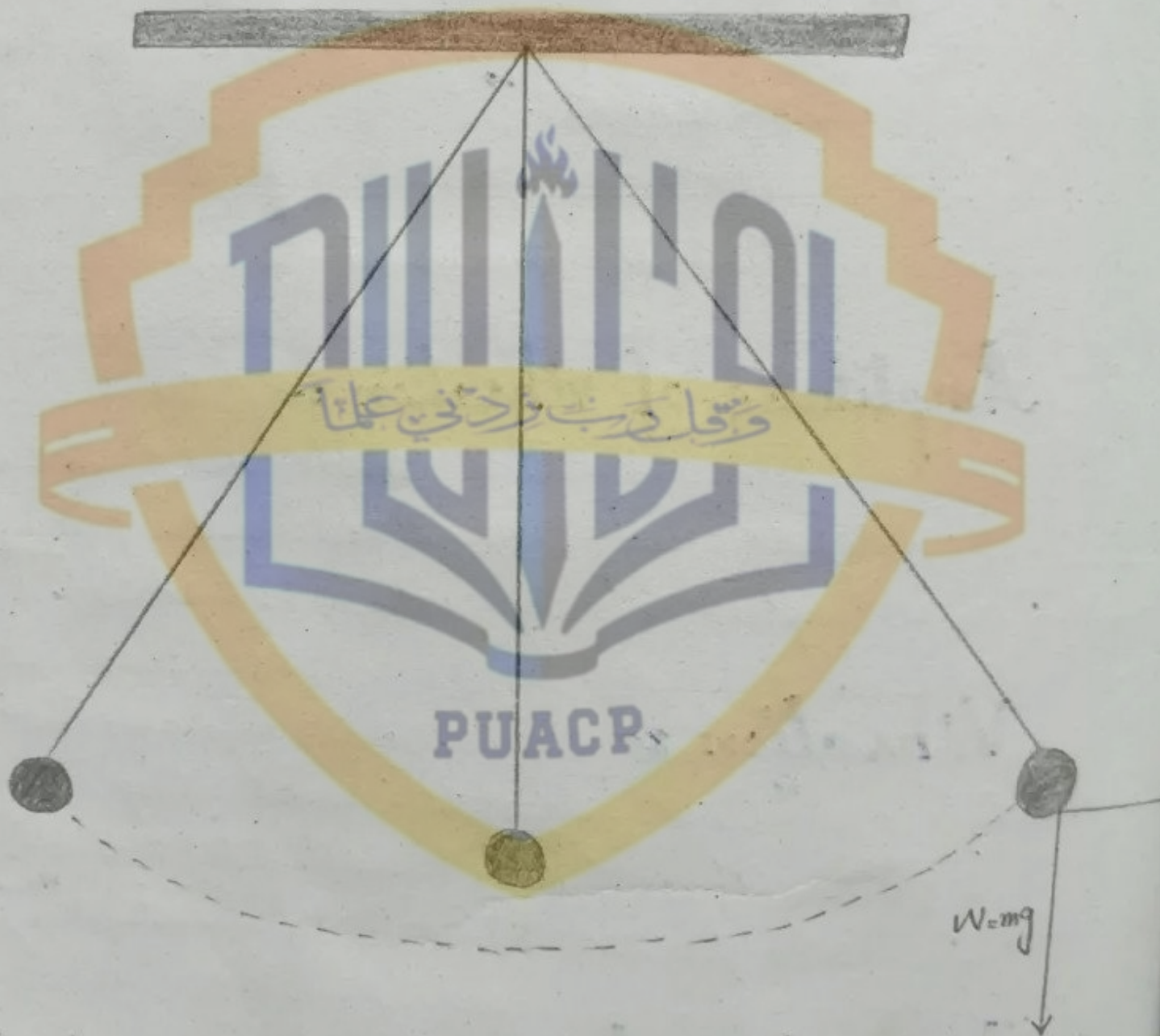
“One complete round trip of a body about its mean position is called a vibration.”

Time Period :-

“Time required to complete 1 vibration is called time Period.”

Units :-

Its units are seconds



$$T = \frac{2\pi}{\omega}$$

$$T = \frac{2\pi}{\sqrt{\frac{k}{m}}}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

It is reciprocal to frequency.

$$T = \frac{1}{f}$$

Frequency :- دقیقہ ارتعاشات

The maximum number of vibration complete by a body in one second is called frequency.

Its units are Hertz or cycle/sec.

$$f = \frac{1}{T}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

Elasticity :-

“The ability of a body to remain to its original shape after the removal of external applied force is called elasticity.”

Stress :-

“The force applied on unit area to produce any change in shape, volume or length of a body.”

Stress measure how strong is material.

Denoted :-

It is denoted by;

$$\sigma = \frac{F}{A}$$

Unit :-

It's units are N/m^2 and also called Pascal's (Pa).

Types :-

The stress has 3 types;

Tensile stress :-

“If the stress change the length of the body is called



tensile or longitudinal stress."

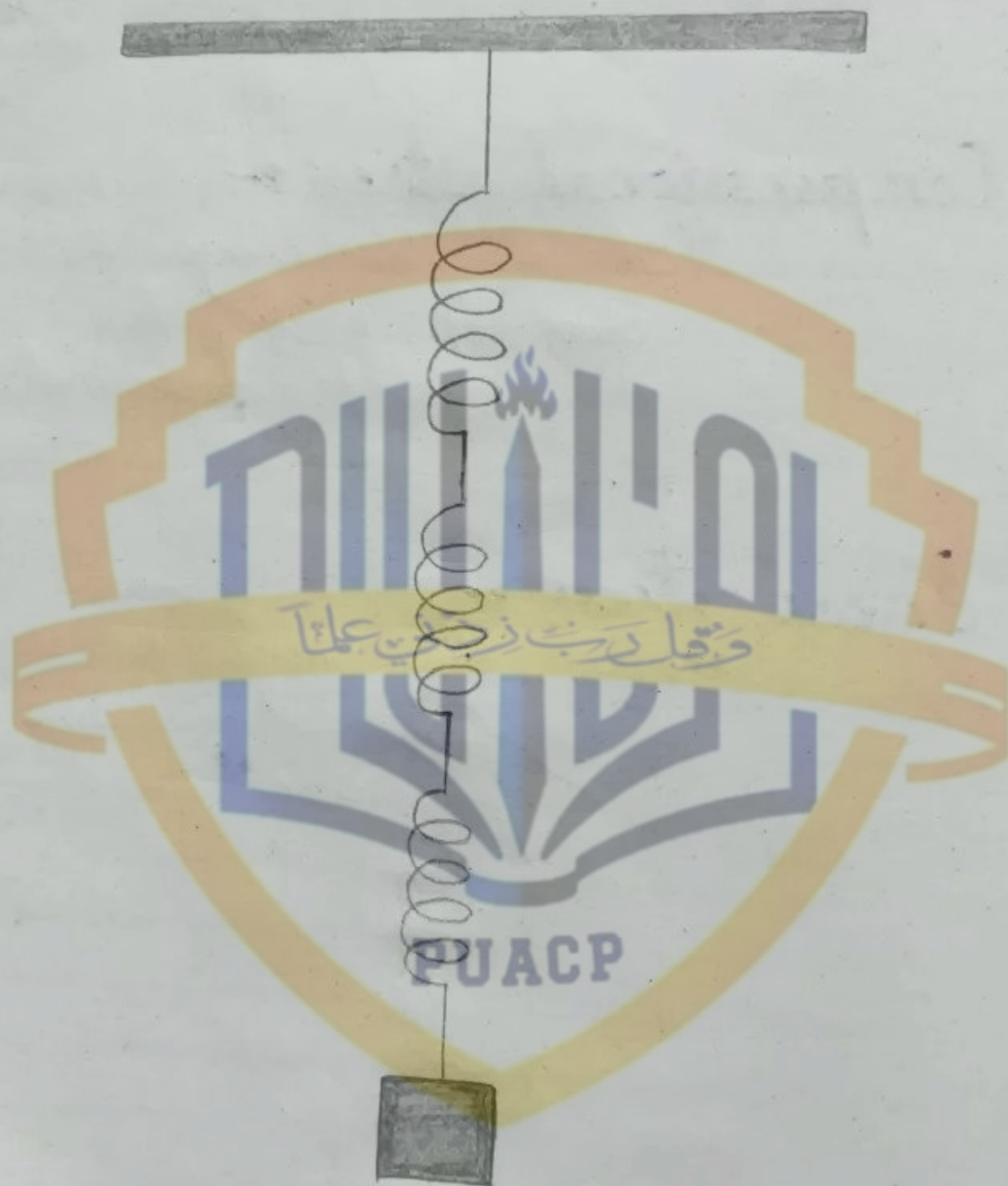
Compressional stress :-

"If a stress change the volume of a body then it is called compressional stress."

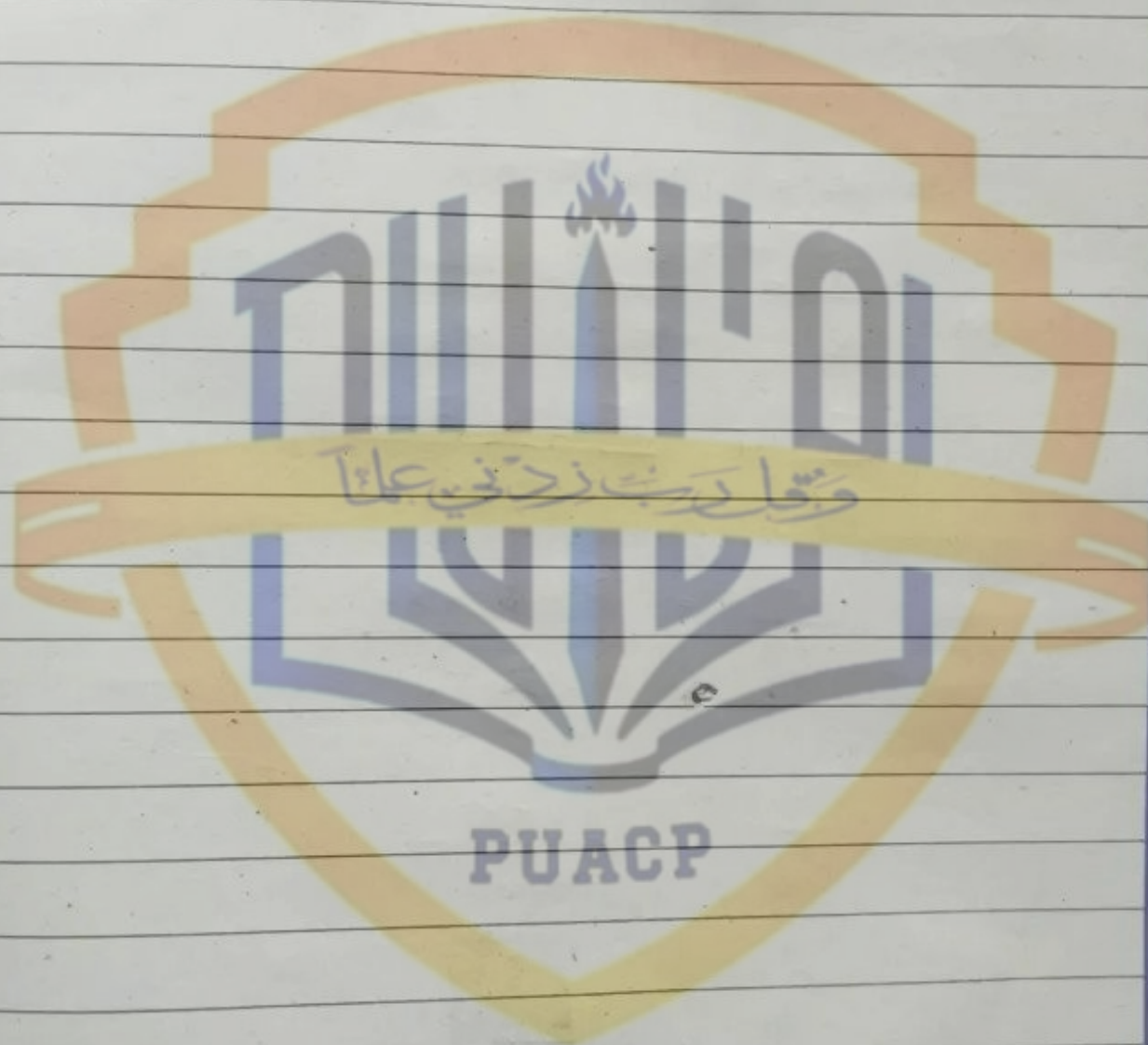
Shear stress :-

"When stress changes the shape of body then it is called shear stress."

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Experiment Name:



$$F = kx$$

$$x = \frac{F}{k}$$

$$F = k_1 x_1$$

$$x_1 = \frac{F_1}{k_1}$$

$$F = k_2 x_2$$

$$x_2 = \frac{F_2}{k_2}$$

$$x = x_1 + x_2$$

$$\frac{F}{k} = \frac{F}{k_1} + \frac{F}{k_2}$$

$$\frac{F}{k} = \frac{Fk_2 + Fk_1}{k_1 k_2}$$

$$\frac{\cancel{F}}{k} = \frac{\cancel{F}(k_2 + k_1)}{k_1 k_2}$$

$$\frac{1}{k} = \frac{k_2 + k_1}{k_1 k_2}$$

$$k = \frac{k_1 k_2}{k_2 + k_1}$$

$$F = k_1 x_1$$

$$x_1 = \frac{F}{k_1}$$

The top supported mg plus weight of bottom spring. F is stretched force.

$$F = k_2 x_2$$

$$x_2 = \frac{F}{k_2}$$

The total stretched:

$$x = x_1 + x_2$$

$$\frac{F}{k} = \frac{F}{k_1} + \frac{F}{k_2}$$

$$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$$

$$k_{eq} = \frac{k_1 k_2}{k_1 + k_2}$$

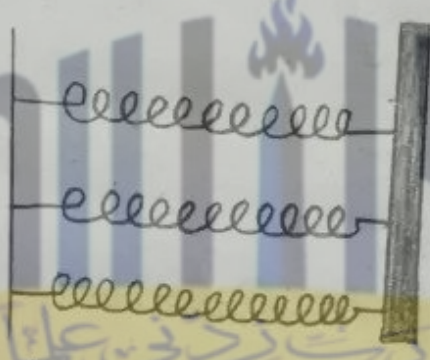
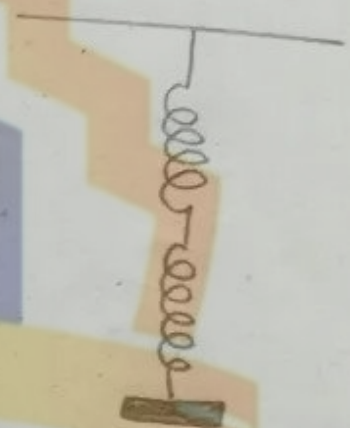
$$k_1 = k_2 = k$$

$$k_{eq} = \frac{k}{2} = k/2$$

$$k_3 = \frac{k_1}{3}$$

$$k_2 = k/2$$

Energy Stored

Comparison	In Parallel	In series
Diagram	 A diagram showing two springs connected in parallel to a horizontal support. A single vertical weight is attached to the bottom of both springs.	 A diagram showing two springs connected in series. The top spring is attached to a horizontal support, and the bottom spring is attached to the bottom of the first spring. A single vertical weight is attached to the bottom of the second spring.
Equivalent Spring Constant	$k_{eq} = k_1 + k_2$	$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$
Compressed Spring	$x_1 = x_2$	$\frac{k_1}{k_2} = \frac{k_2}{k_1}$
Energy Stored	$\frac{E_2}{E_1} = \frac{k_1}{k_2}$	$\frac{E_1}{E_2} = \frac{k_1}{k_2}$

Spring in parallel :-

Two or more spring are said to be parallel when they are connected side by side and act as single spring.

For parallel springs the equivalent spring constant can be calculated as;

$$K_{eq} = k_1 + k_2 + k_3 + \dots + k_n$$

Explanation :-

Consider 2 springs are connected in parallel with constant k_1 and k_2 supported a load $F = mg$

let the force constant of spring combination be represented by k .

For combination support load

$$F = mg.$$

Hook's law :-

$$F = kx$$

the two individual springs both stress by x , but share the load.

For single spring

No. of obs	Mass 'm'	Distance 'x'	Mass 'm'	Distance 'x'	$W = mg$ $W = m(9.8)$	Spring-constant $k = F/x = mg/x$
	g	cm	kg	m	kgms^{-2}	
1	50	6.7	0.05	0.067	0.49	7.32
2	100	13.2	0.1	0.132	0.98	7.42
3	150	17.6	0.15	0.176	1.47	8.35
4	200	22.6	0.2	0.226	1.96	8.67

$$\begin{aligned} \text{Mean Value} &= \frac{7.32 + 7.42 + 8.35 + 8.67}{4} \\ &= \frac{31.76}{4} \\ &= 7.94 \end{aligned}$$

$$F = F_1 + F_2$$

$$F_1 = k_1 x$$

$$F_2 = k_2 x$$

then total force is;

$$F = k_1 x + k_2 x$$

$$k_{eq} = k_1 + k_2$$

Energy stored :-

The P.E of a spring is given by ;

$$P.E = \frac{1}{2} k x^2$$

In Series :-

When two springs are in series the magnitude of the force of the springs on each other are equal.

$$|F_1| = |F_2|$$

$$|k_1 x_1| = |k_2 x_2|$$

$$k_1 x_1 = k_2 (x_2 - x_1)$$

For spring 1, x_1 is the distance from equilibrium length for spring 2 ($x_2 - x_1$) is the distance from its equilibrium position.

So we can write ;

$$a_1 = x_1$$

$$a_2 = x_2$$

For Series

no. of obs.	No. of spring	Time for 15 Vibration			Time Period $T/15$	Time T^2	$k = \frac{4\pi^2 m}{T^2}$
		t_1	t_2	Mean	sec	sec ²	N/m
1.	1	12.82	12.90	12.86	0.857	0.734	8.864
2.	2	17.90	17.59	17.74	1.183	1.396	4.233
3	3	21.34	22.03	21.68	1.44	2.08	2.841

$$k_1 = 8.064$$

$$k_2 = \frac{k_1}{2} = \frac{8.064}{2} = 4.032$$

$$k_3 = \frac{k_1}{3} = \frac{8.064}{3} = 2.688$$

$$\frac{a_1}{a_2} = \frac{k_2}{k_1}$$

For series case of springs, the ratio of energy stored in spring;

$$\frac{E_1}{E_2} = \frac{\frac{1}{2} k_1 x_1^2}{\frac{1}{2} k_2 x_2^2}$$

$$\frac{E_1}{E_2} = \frac{k_1}{k_2} \cdot \frac{x_1^2}{x_2^2}$$

We know that $x_1 = a_1$ and $x_2 = a_2$

$$\frac{E_1}{E_2} = \frac{k_1}{k_2} \cdot \frac{a_1^2}{a_2^2}$$

$$\frac{E_1}{E_2} = \frac{k_1}{k_2} \left(\frac{k_2}{k_1} \right)^2$$

$$\frac{E_1}{E_2} = \frac{k_2}{k_1}$$

In Parallel:-

For parallel case, the distance of both spring from the equilibrium position is same so;

$$x_1 = x_2$$

And ratio of energy;

$$\frac{E_1}{E_2} = \frac{\frac{1}{2} k_1 x_1^2}{\frac{1}{2} k_2 x_2^2}$$

For Parallel

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No. of obs.	No. of spring	Time for 15 Vibration			Time Period		$k = \frac{4\pi^2 m}{T^2}$ N/m
		t_1	t_2	Mean	$T/15$ (sec)	T^2 sec ²	
1.	2	8.86	8.70	8.76	0.584	0.341	16.32
2.	3	7.47	7.32	7.395	0.493	0.2423	24.43

$$k_1 = 8.064$$

$$k_2 = 2k_1 = 2(8.064) = 16.128$$

$$k_3 = 3k_1 = 3(8.064) = 24.192$$

Since the distance is same;

$$\frac{E_1}{E_2} = \frac{k_1}{k_2}$$

Procedure:-

- ◆ Take an iron stand and add spring with its hanger.
- ◆ Add weight and spring stretched some distance.
- ◆ Note the distance from metal rod.
- ◆ Add more weight and note the distance x .
- ◆ We can calculate spring constant by using formula;

$$F = kx$$

$$k = \frac{F}{x}$$

$$k = \frac{mg}{x}$$

- ◆ By hang 2 spring in series and parallel.
- ◆ Note time vibration for 20 vibration.
- ◆ Find time period similarly repeat process for 3 time.
- ◆ For parallel connection also find time period and k by using formula;

$$k_1 = 8.064$$

No. of spring	Combination of Spring	Spring constant - k (N/m)		
		Theoretical Value	Experimental Value	Error
1		7.94	8.064	1.5%
2	For Series	$k_2 = \frac{k_1}{2}$ 4.032	4.2337	2.5%
3	For Series	$k_3 = \frac{k_1}{3}$ 2.688	2.8471	2.0%
2	For Parallel	$k_2 = 2k_1$ 16.12	16.32	2.5%
3	For Parallel	$k_3 = 3k_1$ 24.29	24.43	3%

$$\text{Percentage error} = \frac{8.084 - 7.94}{7.94} \times 100$$

$$= 1.5\%$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

Squaring both sides:

$$T^2 = \frac{4\pi^2 m}{k}$$

$$k = \frac{4\pi^2 m}{T^2}$$

- ◆ In the end we will compare theoretical and actual value.

Precautions:-

- ◆ The stand should be heavy and stable.
- ◆ Spring should hang freely.
- ◆ Take reading when pointer come to rest.
- ◆ Spring should not be located beyond the elastic limit.
- ◆ Take the level in front of eye.

$$k_2 = 2k_1$$

$$k_3 = 3k_1$$